

Graph Domain Adaptation: A Detailed Survey of Representation, Structural, Spectral, Causal, and Continual Adaptation

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Abstract

Graph domain adaptation (GDA) studies how knowledge learned from one or more source graphs can be transferred to a target graph whose data distribution differs and whose labels are limited or unavailable. The problem has become increasingly important as graph neural networks (GNNs) are deployed across social networks, recommendation systems, fraud detection, biological interaction networks, transportation systems, and scientific graphs. However, unlike conventional domain adaptation on independent samples, GDA must cope with intertwined shifts in node attributes, graph topology, neighborhood statistics, label proportions, homophily, spectral properties, and even the label space itself. This makes naive feature alignment insufficient and raises fundamental questions about what graph knowledge is transferable, how transferability should be measured, and how source knowledge should be adapted without causing negative transfer. This survey develops a unified, operation-oriented view of GDA. We first formulate the problem at node, edge, graph, and graph-set levels, and distinguish homogeneous, heterogeneous, closed-set, open-set, source-free, multi-source, and continual settings. We then organize the literature into six major adaptation mechanisms: (1) representation and distribution alignment, (2) structural and topology alignment, (3) propagation, homophily, and spectral adaptation, (4) prototype, contrastive, and pseudo-label based adaptation, (5) causal and generative adaptation, and (6) multi-source, source-free, and continual adaptation. A second orthogonal axis describes where adaptation is injected: the source data/model, the adaptation objective, or the target graph. This perspective reveals that many apparently different methods implement recurring operations such as normalize, align, reweight, propagate, intervene, filter, aggregate, and memorize. Beyond summarizing methods, the survey emphasizes the assumptions behind each strategy, the failure modes they introduce, the datasets and evaluation protocols used in the field, and the gaps that remain between benchmark performance and real-world deployment. We discuss the importance of homophily mismatch, structural heterogeneity, negative transfer, target label uncertainty, source selection, model calibration, privacy constraints, and temporal drift. We also review major developments through 2025–2026, including homophily-aware GDA, source-free multi-source adaptation, causal generative intervention, spectral disentanglement, open-set graph adaptation, graph data selection, and adaptation to evolving domain shifts. We conclude with a research agenda centered on transferability estimation, causal invariance, scalable structural matching, uncertainty-aware adaptation, foundation models for graphs, and unified benchmarks for realistic graph distribution shifts.

Keywords: graph domain adaptation; graph neural networks; transfer learning; graph representation learning; unsupervised domain adaptation; domain shift; structural adaptation; homophily; source-free adaptation; continual graph learning

1. Introduction

Machine learning pipelines are usually developed under a simplifying assumption: the data available during training and the data encountered during deployment are drawn from similar

distributions. Domain adaptation challenges this assumption by studying the transfer of knowledge from a labeled source domain to a related but distributionally different target domain. In conventional settings, examples are frequently treated as independent observations, and domain shift is described through differences in feature distributions, conditional label distributions, or class priors.

Graphs fundamentally change this picture. A graph encodes both attributes and relations, and the prediction for one node can depend on its neighbors, its local topology, the distribution of its community, or even long-range structural patterns. A social network, a citation graph, a molecular graph, a recommender interaction graph, and a road network all exhibit this dual nature: data are not only points, but points embedded in a relational structure. When the graph domain changes, several sources of shift can occur simultaneously. Features may change, edges may be rewired, degree distributions may differ, communities can appear or disappear, homophily can change, and the mapping from attributes and structure to labels can drift.

Graph domain adaptation (GDA) has therefore emerged as a specialized area at the intersection of graph representation learning and transfer learning. The first comprehensive survey devoted specifically to GDA appeared in 2025 and organized the field around the source, adaptation mechanism, and target sides of the transfer pipeline [1]. Since then, the literature has expanded toward structurally aware methods, causal adaptation, source-free and multi-source settings, open-set adaptation, homophily alignment, data-centric selection, and continual adaptation.

The central challenge is not merely to make source and target representations similar. A useful adaptation mechanism must preserve information that remains predictive across domains while discarding domain-specific variation that is likely to be spurious. This is difficult because the representation that is invariant at one abstraction level may be non-invariant at another. For instance, node features may be aligned while neighborhoods remain structurally incompatible. Conversely, two graphs can have different observed features but nearly identical spectral or class-conditional structural behavior.

This survey is designed as a detailed, system-oriented reference rather than a chronological list of papers. Its organizing principle is that GDA methods repeatedly perform a small set of operations on different graph objects. A method may transform source data, modify the adaptation objective, reweight source models, alter the target graph, or memorize information across domains. The resulting taxonomy makes it easier to compare approaches that use different mathematical machinery but solve closely related transfer problems.

1.1. Contributions of the Survey

The survey makes six organizing contributions.

1. It provides a unified mathematical formulation of GDA across node-level, graph-level, heterogeneous, open-set, source-free, multi-source, and continual settings.
2. It proposes a two-axis taxonomy: *what is adapted* (representation, structure, spectrum/propagation, prediction, causality, or memory) and *where adaptation acts* (source, adaptation objective, or target).
3. It synthesizes the evolution of the field from early graph spectral transfer and cross-network embeddings to modern GNN, contrastive, causal, and continual approaches.
4. It analyzes common assumptions and failure modes, including negative transfer, homophily mismatch, structural heterogeneity, pseudo-label error, open-set contamination, and catastrophic forgetting.
5. It reviews evaluation practices, benchmark construction, transfer scenarios, and methodological pitfalls that can make comparisons misleading.
6. It develops a forward-looking research agenda focused on transferability estimation, causal invariance, uncertainty, scalability, graph foundation models, and realistic benchmarks.

1.2. Relationship to Existing Surveys

The 2025 survey by Shi et al. provides the first broad overview of graph domain adaptation and identifies source-, adaptation-, and target-oriented categories [1]. The present survey extends that taxonomy in two directions. First, it treats homophily, spectral filtering, and propagation as a distinct structural-signal family rather than subsuming them under generic representation alignment. Second, it gives dedicated treatment to post-2025 developments in source-free multi-source GDA, continual adaptation, causal intervention, graph data selection, and open-set adaptation.

2. Background: Graph Learning and Domain Adaptation

2.1. Graph Representation Learning

Let a graph be $G = (V, E, X)$, where V denotes nodes, E denotes edges, and $X \in \mathbb{R}^{|V| \times d}$ contains node attributes. Its adjacency matrix is $A \in \mathbb{R}^{|V| \times |V|}$. A message-passing graph neural network computes node representations iteratively:

$$H^{(\ell+1)} = \sigma(\text{AGG}^{(\ell)}(H^{(\ell)}, A)W^{(\ell)}), \quad (1)$$

with $H^{(0)} = X$. In a common GCN form,

$$H^{(\ell+1)} = \sigma(\hat{A}H^{(\ell)}W^{(\ell)}), \quad (2)$$

where $\hat{A}$ is a normalized adjacency matrix. GNNs learn representations by jointly exploiting feature information and graph structure, but this also makes them sensitive to distribution shifts in either component.

2.2. Conventional Domain Adaptation

In conventional unsupervised domain adaptation, the source domain contains labeled samples (x_i^s, y_i^s) and the target domain contains unlabeled samples x_j^t . A common objective is

$$\mathcal{L} = \mathcal{L}_{\text{src}} + \lambda \mathcal{L}_{\text{align}}, \quad (3)$$

where $\mathcal{L}_{\text{src}}$ is a source supervised loss and $\mathcal{L}_{\text{align}}$ reduces a discrepancy between source and target distributions. MMD, adversarial discriminators, optimal transport, entropy minimization, and pseudo-labeling are representative mechanisms.

GDA inherits these ideas but must additionally account for relational dependency and graph topology. This is why the discrepancy should often be defined over graph-conditioned representations rather than independent samples.

3. Unified Problem Formulation

3.1. Source and Target Graphs

Let the source graph and target graph be

$$G_s = (V_s, E_s, X_s, Y_s), \quad G_t = (V_t, E_t, X_t, Y_t). \quad (4)$$

For node classification, Y_s is observed while Y_t is hidden during training. The goal is to learn f_θ that minimizes target risk

$$\mathcal{R}_t(f_\theta) = \mathbb{E}_{(x,a,y) \sim P_t}[\ell(f_\theta(x,a), y)]. \quad (5)$$

Because target labels are missing, training minimizes a source objective plus one or more transfer objectives.

3.2. What Can Shift?

Graph domain shift is multidimensional. We distinguish at least seven major forms.

Covariate shift.

$P_s(X) \neq P_t(X)$ while the label mechanism remains approximately invariant.

Structural shift.

$P_s(A) \neq P_t(A)$, including differences in density, degree distribution, motif frequency, community structure, or graph size.

Conditional shift.

$P_s(Y | X, A) \neq P_t(Y | X, A)$, meaning that the same observed attributes or structural patterns can imply different labels.

Label-prior shift.

$P_s(Y) \neq P_t(Y)$, which can invalidate global alignment when class proportions differ substantially.

Homophily shift.

The tendency for adjacent nodes to share labels differs between source and target graphs. Recent work demonstrates that homophily mismatch can materially degrade GDA performance [10].

Spectral shift.

The frequency content associated with useful label functions differs across graph Laplacians or graph filters. Spectral alignment methods were among the earliest graph-specific adaptation approaches [2].

Label-space shift.

The target domain may include classes absent from the source domain, yielding the open-set GDA problem [7,16].

3.3. Task Granularity

GDA can be formulated at four granularity levels.

Table 1. Graph domain adaptation settings by task granularity.

Level	Description	Typical tasks
Node	Adapt representations or predictions for nodes across graphs.	Node classification, role prediction, anomaly detection.
Edge	Transfer information for pairwise relations.	Link prediction, interaction prediction, edge classification.
Graph	Treat each graph as an instance.	Graph classification, regression, property prediction.
Graph-set	Adapt across distributions of many graphs.	Multi-domain graph learning, molecular or scientific transfer.

3.4. Closed-Set, Open-Set, and Partial Settings

Classical UDA assumes a shared label space. Open-set GDA relaxes this by allowing unknown target classes. In a closed-set setting,

$$\mathcal{Y}_s = \mathcal{Y}_t. \quad (6)$$

In an open-set setting,

$$\mathcal{Y}_s \subsetneq \mathcal{Y}_t, \quad (7)$$

and the system must identify unknown target categories while preserving accuracy on shared categories [7,16].

3.5. Single-Source, Multi-Source, and Source-Free Settings

Single-source GDA uses one labeled graph. Multi-source GDA introduces K labeled source graphs:

$$\mathcal{S} = \{G_1^s, \dots, G_K^s\}. \quad (8)$$

The objective is then to learn a target representation from complementary and potentially conflicting sources. Recent multi-source work introduces node-level and domain-level transferability weights instead of treating all sources equally [18].

Source-free GDA assumes that the original labeled source graph is no longer available, and only a pretrained source model, together with unlabeled target data, can be used for adaptation. This setting is attractive under privacy, regulatory, and storage constraints and has motivated model-only approaches such as GraphATA [15].

4. A Unified Taxonomy of Graph Domain Adaptation

The field can be organized around the following pipeline:

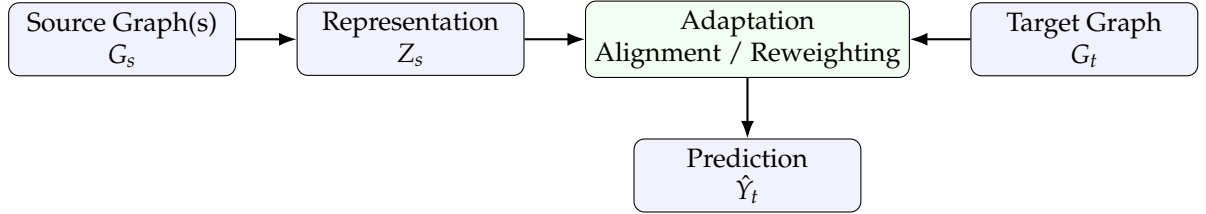


Figure 1. A unified view of graph domain adaptation. Different methods intervene at the source, representation, adaptation objective, or target graph.

We define six broad method families.

1. **Representation alignment:** align source and target embeddings or distributions.
2. **Structural, homophily, and spectral adaptation:** align graph signals, topology, neighborhoods, or frequency behavior.
3. **Prediction-aware adaptation:** use prototypes, pseudo-labels, entropy, or contrastive objectives to align class-conditional structure.
4. **Causal and generative adaptation:** separate transferable mechanisms from spurious factors or model the domain transformation explicitly.
5. **Data-centric and source-centric adaptation:** select, reweight, aggregate, or transform source data and source models.
6. **Continual and memory-based adaptation:** maintain transferability over sequences of evolving target domains.

This taxonomy is orthogonal to the source/adaptation/target categorization used in the 2025 survey [1]. A method can therefore be described along both axes. For example, TDSS primarily acts on the target structural representation, while ACDNE operates through the adaptation objective, and GraphATA combines source-model aggregation with target-aware node-level adaptation.

5. Methodological Lineage from Graph Matching and Classical Domain Adaptation

A useful way to understand modern GDA is to connect graph-specific methods to two important methodological lineages: (i) graph and hyper-graph matching for finding sample correspondences across domains, and (ii) discrepancy, reweighting, regularization, and adversarial methods developed in classical domain adaptation. This connection is particularly important because several ideas that are now standard in GDA have close conceptual precedents in work by Das and Long.

5.1. Das: Graph Matching as an Adaptation Primitive

Das and Lee proposed a family of unsupervised domain adaptation methods in which source and target samples are explicitly represented as graphs and matched using structural similarity. Their 2018 work on sample-to-sample correspondence uses first- and second-order graph similarities together with class-based regularization to identify correspondences between domains [20]. This is a direct precursor to modern structure-aware GDA: rather than assuming that two domains can be aligned adequately through global moments, it attempts to identify which source and target instances correspond locally.

The hyper-graph extension strengthens this idea by representing higher-order relations and performing class-regularized matching using first-, second-, and third-order similarities [21]. The resulting optimization can be interpreted as a structural transport mechanism. In modern GDA language, this anticipates several recurring operations:

$$\begin{aligned} &\text{construct relational similarity} \rightarrow \text{match cross-domain structure} \\ &\rightarrow \text{transfer information through correspondences.} \end{aligned} \quad (9)$$

Das and Lee also introduced a graph matching and pseudo-label guided deep unsupervised domain adaptation formulation [22]. The important conceptual step is that pseudo-label information is not used only after alignment; it helps determine which structural correspondences should be trusted. This creates a feedback loop between prediction and structure, a pattern that later appears in prototype alignment, class-conditional adaptation, and pseudo-label refinement in graph neural networks.

A second contribution of this lineage is the explicit treatment of temporal adaptation. Moon, Das, and Lee formulated online unsupervised domain adaptation in which unlabeled target samples arrive sequentially. Their multi-step framework maintains accumulated target statistics and uses a mean-target subspace so that information from earlier target batches informs subsequent adaptation [23,24]. Although these papers are not graph-neural network papers, their treatment of sequential target drift is directly relevant to continual GDA, where the target graph evolves instead of appearing as a single static object.

5.2. Long: The Statistical and Adversarial Foundations

Long’s work provides several of the core mechanisms that modern GDA methods inherit. Joint Distribution Adaptation (JDA) introduced the idea of aligning both marginal and conditional distributions rather than matching only low-order feature statistics [27]. This distinction is especially important for graphs because a representation can be globally aligned while remaining class-inconsistent. A graph-aware version of the same principle can be written as

$$\mathcal{L}_{\text{align}} = \mathcal{D}(P_s(Z), P_t(Z)) + \beta \mathcal{D}(P_s(Z | Y), P_t(Z | Y)), \quad (10)$$

where the second term prevents samples from different semantic classes from being indiscriminately merged.

Transfer Joint Matching (TJM) then coupled feature matching with instance reweighting [28]. This combination is particularly relevant to GDA because a source graph may contain nodes that have no reliable counterpart in the target graph. Structural alignment should therefore often be selective

rather than uniform. In graph terms, node importance, edge importance, or subgraph relevance can be treated as transfer weights.

Adaptation Regularization (ARTL) unified structural risk minimization, joint-distribution matching, and manifold consistency in a single transfer-learning objective [29]. This formulation foreshadows modern GDA objectives that combine supervised source loss with domain discrepancy and graph-smoothness terms. Graph Co-Regularized Transfer Learning (GTL) makes the connection even more explicit by jointly exploiting statistical information and geometric structure to extract common latent factors [25,26].

Deep Adaptation Networks (DAN) generalized kernel mean matching to task-specific deep representations using multi-kernel maximum mean discrepancy (MK-MMD) [30]. Joint Adaptation Networks (JAN) extended this idea from marginal feature alignment to joint alignment across multiple domain-specific layers using joint MMD [31]. Conditional Adversarial Domain Adaptation (CDAN) further argued that adversarial adaptation should be conditioned on discriminative classifier information so that multimodal class structure is not destroyed [32].

For GDA, these developments suggest a progression:

$$\begin{aligned}
 &\text{marginal alignment} \rightarrow \text{joint/class-conditional alignment} \\
 &\rightarrow \text{conditional adversarial alignment} \\
 &\rightarrow \text{structure-aware conditional alignment.}
 \end{aligned} \tag{11}$$

The last step is not a direct claim made in the classical papers; it is the natural graph-specific extension developed by subsequent GDA research.

Finally, Long and collaborators developed theory connecting domain discrepancies, margin-based risks, and adversarial optimization, including Margin Disparity Discrepancy and localized discrepancy [33,34]. These results matter for GDA because they clarify that a discrepancy is useful only when it is coupled to the hypothesis class and task loss. This provides a theoretical argument against indiscriminate graph alignment.

5.3. How These Lineages Map onto Modern GDA

Table 2 summarizes the conceptual correspondence between these foundational contributions and graph-domain adaptation operations.

Table 2. Foundational adaptation mechanisms and their role in modern GDA.

Lineage	Core operation	GDA interpretation
Das–Lee	Graph / hyper-graph matching	Explicit structural correspondences between source and target nodes or local neighborhoods.
Das–Lee	Pseudo-label guided matching	Jointly refine structural alignment and semantic predictions.
Das–Lee / Moon	Sequential target statistics	Adapt to temporal or evolving target domains without resetting the model.
Long et al.	Joint distribution adaptation	Align marginal and class-conditional representations.
Long et al.	Joint feature matching + reweighting	Down-weight irrelevant or non-transferable source nodes.
Long et al.	Graph co-regularization	Combine statistical transfer with geometric or manifold consistency.
Long et al.	MK-MMD / JMMD	Align distributions at one or multiple representation levels.
Long et al.	Conditional adversarial adaptation	Preserve multimodal class structure during domain alignment.
Long et al.	Adaptation theory	Connect discrepancy choices to transfer risk and generalization.

5.4. Implications for the GDA Taxonomy

These works motivate an important refinement of the survey taxonomy. Domain adaptation should not be organized only by the loss function used to make two distributions similar. A more complete view distinguishes at least four questions:

1. **What is aligned?** Features, class conditionals, nodes, edges, neighborhoods, spectra, prototypes, or causal factors.
2. **How is alignment weighted?** Uniformly, by instance importance, by class confidence, by structure, or by source-level transferability.
3. **Where does supervision enter?** Source labels, pseudo-labels, prototypes, graph propagation, or temporal memory.
4. **When is adaptation applied?** Once, iteratively, sequentially, or continuously under domain evolution.

This four-question view helps place both the Das–Lee and Long research programs inside the broader GDA literature without treating them as isolated historical references.

6. Representation Alignment

Representation alignment is the most direct extension of conventional domain adaptation to GNNs and follows the discrepancy-matching lineage developed by Long et al. through JDA, DAN, and JAN [27,30,31]. The idea is to learn a shared embedding space in which source and target representations are difficult to distinguish.

6.1. Moment and Kernel Alignment

Suppose Z_s and Z_t are latent node representations. A kernel-based discrepancy may be written as

$$\mathcal{L}_{\text{MMD}} = \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \phi(z_i^s) - \frac{1}{n_t} \sum_{j=1}^{n_t} \phi(z_j^t) \right\|^2. \quad (12)$$

Such objectives are simple and stable, but they generally align marginal distributions and can overlook class structure. A global match can map nodes from different classes into similar regions, especially when source and target priors differ.

6.2. Adversarial Alignment

Adversarial methods introduce a domain discriminator D_ψ and train the encoder to hide domain identity:

$$\min_{\theta} \max_{\psi} \mathcal{L}_{\text{task}}(\theta) - \lambda \mathcal{L}_{\text{dom}}(\theta, \psi). \quad (13)$$

ACDNE combines domain adversarial learning with structural and attribute-based network embedding for cross-network node classification [3]. The main advantage is that the discrepancy function is learned. The main risk is over-alignment: if the domains have genuinely different label mechanisms, a strong discriminator objective can suppress useful domain-specific information.

6.3. Target-Oriented Alignment

TO-UGDA explicitly emphasizes target-oriented adaptation rather than purely symmetric alignment [5]. The broader principle is that the target domain is the deployment objective, so alignment should be weighted by target relevance instead of being treated as a generic distance minimization problem.

6.4. Key Insight

Representation alignment is necessary in many settings but rarely sufficient on its own. Later work repeatedly identifies the same limitation: if alignment ignores structure, homophily, or class-

conditional information, a model can achieve low domain discrepancy while retaining high target error [6,9,10].

Takeaway. Representation alignment is best viewed as a baseline operation that requires an additional mechanism for deciding *which* representations should be aligned and *under what structural conditions*.

7. Structural and Topological Adaptation

Graph structure is not a nuisance variable; it directly controls message passing. Structural adaptation therefore targets the adjacency matrix, neighborhood distribution, motifs, communities, or structural embeddings.

7.1. Graph Basis Alignment

Pilanci and Vural proposed one of the earliest graph-specific approaches by learning aligned graph Fourier bases between source and target graphs [2]. The key idea is to transfer the spectral characteristics of the label function rather than directly transferring node-level features. Given Laplacians L_s and L_t with eigenvectors U_s and U_t , one can learn a transformation T such that

$$U_s \approx U_t T. \quad (14)$$

The method is important conceptually because it demonstrates that domain adaptation can occur entirely in graph signal space without embedding the data in a conventional ambient feature space.

7.2. Structure-Attribute Joint Alignment

Recent work increasingly treats structure and attributes jointly. SATMC models graph transformation through a sequence that modifies structure and then attributes, using a Markov-chain perspective and explicit reduction of private information [17]. This addresses an important limitation of methods that keep the target topology fixed while attempting to align only the node features.

7.3. Structural Smoothing

TDSS argues that small structural differences can induce significant changes in GNN embeddings. Instead of directly matching graph structures, it performs target-domain structural smoothing with neighborhood sampling to improve representation consistency while limiting over-smoothing [9].

7.4. Why Structure Matters

Two graphs can have nearly identical node feature histograms but radically different predictive neighborhoods. Consider a citation network where topic words are stable but citation conventions differ across research communities. A node’s neighbors may carry much of the label information, so feature alignment without topology adaptation can be misleading.

Table 3. Examples of structural mismatch and corresponding adaptation strategies.

Shift	Observed effect	Typical response
Density shift	Message aggregation changes in magnitude and variance.	Normalization, structural reweighting, adaptive aggregation.
Degree shift	High-degree nodes dominate representation propagation.	Degree-aware weighting, sampling, reparameterization.
Community shift	Neighborhoods encode different semantics.	Structural alignment, sub-graph alignment.
Homophily shift	Neighbor labels become less informative or differently distributed.	Homophily alignment, smoothing/filter adaptation.
Spectral shift	Useful label frequencies differ across domains.	Graph Fourier or spectral filter adaptation.

Takeaway. Structure should be treated as a first-class adaptation variable rather than a fixed container around feature adaptation.

8. Propagation, Homophily, and Spectral Adaptation

This section groups methods that explicitly reason about the graph signal-processing behavior induced by message passing.

8.1. Propagation as an Adaptation Mechanism

Liu et al. show that the propagation process itself can play a central role in unsupervised GDA and argue that adaptation should not be reduced to latent-space alignment [6]. In a message-passing system, the propagation operator determines how information from neighboring nodes is mixed. If source and target graphs differ, the same parameter matrix W can induce different representations purely because $A_s \neq A_t$.

This motivates separating transformation and propagation. Instead of applying a single entangled GNN block, one can learn a feature transformation that is domain-aware while using a target-sensitive propagation operator. This idea reappears in multi-source methods such as PMGCDA [18].

8.2. Homophily Alignment

Homophily is the tendency of connected nodes to share labels or semantic attributes. It is central to the success of neighborhood aggregation. If the source graph is highly homophilous and the target graph is weakly homophilous, a GNN trained on the source can over-trust neighbors in the target.

Fang et al. provide both empirical and theoretical evidence that homophily discrepancies in benchmarks harm GDA performance and propose mixed graph filters to align homophily across domains [10]. This is significant because it shifts the adaptation objective from matching embeddings to matching a structural property that directly controls message-passing quality.

8.3. Spectral Adaptation

Graph signal processing offers a complementary perspective. Let $L = U\Lambda U^\top$ be a graph Laplacian eigendecomposition. A spectral graph filter can be written as

$$H' = Ug(\Lambda)U^\top H, \quad (15)$$

where g is a learnable or predefined frequency response. Domain adaptation can then operate by learning filters whose responses are stable across source and target graphs.

DGSDA explicitly disentangles attribute and topology alignment and directly adapts graph spectral filters instead of relying only on topology estimation and pseudo-labels [12].

Takeaway. Propagation, homophily, and spectral characteristics form an intermediate layer between raw graph structure and learned embeddings. They offer a promising language for explaining why a GNN transfers or fails.

9. Prototype, Contrastive, and Prediction-Aware Adaptation

Global distribution alignment can be blind to class structure. Prediction-aware methods therefore use pseudo-labels, prototypes, entropy, or contrastive objectives to align source and target instances conditionally.

9.1. Class-Conditional Alignment

Let c denote a class. Instead of aligning $P_s(Z)$ with $P_t(Z)$, one can align

$$P_s(Z \mid Y = c) \approx P_t(Z \mid Y = c). \quad (16)$$

When target labels are unavailable, pseudo-labels $\tilde{Y}_t$ or class prototypes are used.

9.2. Prototype-Based Adaptation

A class prototype can be defined as

$$\mu_c^s = \frac{1}{|V_c^s|} \sum_{i: y_i^s = c} z_i^s, \quad \mu_c^t = \frac{1}{|\tilde{V}_c^t|} \sum_{j: \tilde{y}_j^t = c} z_j^t. \quad (17)$$

An alignment loss is then

$$\mathcal{L}_{\text{proto}} = \sum_c \|\mu_c^s - \mu_c^t\|_2^2. \quad (18)$$

The risk is confirmation bias: incorrect pseudo-labels produce incorrect prototypes, which can reinforce the wrong decision boundary.

9.3. Contrastive Adaptation

Contrastive objectives improve class separation while reducing same-class cross-domain discrepancy. A generic loss can be written as

$$\mathcal{L}_{\text{con}} = - \sum_i \log \frac{\exp(\text{sim}(z_i, z_i^+) / \tau)}{\exp(\text{sim}(z_i, z_i^+) / \tau) + \sum_{j \in \mathcal{N}_i} \exp(\text{sim}(z_i, z_j) / \tau)}, \quad (19)$$

where z_i^+ is a positive cross-domain example and $\mathcal{N}_i$ contains negatives.

The 2026 PMGCDA method extends this idea to multi-source adaptation through local and global prototypes, using node-level and domain-level transferability weights to control how much each source contributes to each target node [18].

9.4. Entropy and Prediction Consistency

Unsupervised target entropy minimization can be written as

$$\mathcal{L}_{\text{ent}} = - \frac{1}{|V_t|} \sum_{i \in V_t} \sum_c p_{ic} \log p_{ic}. \quad (20)$$

The goal is to produce confident target predictions. Confidence, however, is not equivalent to correctness. Entropy minimization is therefore safer when combined with confidence thresholds, consistency regularization, or structural constraints.

Takeaway. Prediction-aware adaptation moves GDA from global domain matching toward class-aware transfer. Its success depends heavily on the quality of pseudo-labels and prototypes.

10. Causal and Generative Adaptation

A growing body of work argues that the central issue in domain adaptation is not discrepancy but causality. Some attributes and structures are stable causes of the label, while others are domain-specific correlates.

10.1. Causal View of GDA

Let Z_c denote causal or invariant features and Z_s denote spurious domain-dependent features. An ideal representation satisfies

$$P_s(Y | Z_c) = P_t(Y | Z_c), \quad (21)$$

while allowing

$$P_s(Z_s) \neq P_t(Z_s). \quad (22)$$

The goal is therefore not to eliminate every domain difference but to isolate the subset that changes the label mechanism.

10.2. Sparse Causal Discovery with Generative Intervention

SLOGAN uses sparse causal discovery and generative intervention to disentangle stable causal features from domain-dependent spurious correlations [11]. This is an important departure from purely discriminative alignment because the method explicitly asks which graph variables should remain invariant.

10.3. Generative Domain Transformation

A generative view models target graphs as transformations of source-like latent factors. Conceptually,

$$G_t \sim p_\theta(G | Z_{\text{shared}}, Z_{\text{private}}^t). \quad (23)$$

Cai et al. formalize GDA through a generative perspective, highlighting the possibility of learning shared latent factors while preserving domain-specific components [19].

10.4. Related Cross-Domain Few-Shot and Test-Time Adaptation Work

Although the main focus of this survey is graph domain adaptation, several lines of non-graph domain adaptation research provide useful methodological context. Das and Lee’s two-stage few-shot framework separates transferable feature extraction from category-level adaptation and introduces relative features and class-variance modeling [35]. Their later ConFeSS framework addresses large shifts between base and novel categories in single-source cross-domain few-shot learning through contrastive representation learning and feature selection [36]. These ideas are relevant to GDA when target graphs contain only a small number of labeled nodes or when source and target label semantics are only partially aligned.

Das and Lee also studied zero-shot recognition using relational matching, test-time adaptation, and score calibration [37]. The emphasis on relational correspondence and target-time adaptation is particularly relevant to graph settings, where relationships rather than isolated observations carry much of the predictive signal. Their manifold-based few-shot formulation further models class prototypes as structured points on a manifold and uses graph-based distances for recognition [38], offering a useful conceptual bridge between few-shot recognition and graph-based prototype adaptation.

Finally, Lee et al. investigated open-set test-time adaptation under entropy minimization and showed that incorrect and open-set predictions can generate noisy adaptation signals; their “wisdom of crowds” filtering strategy selects samples whose confidence does not deteriorate during adaptation [39]. While the method is developed for visual tasks, its emphasis on noisy pseudo-label filtering

and open-set contamination is directly relevant to graph test-time adaptation and source-free GDA, particularly when target neighborhoods contain previously unseen classes. These works are therefore included as supporting rather than core graph references.

10.5. Advantages and Limitations

Causal and generative methods provide richer explanations of transferability, but their objectives can be difficult to identify from observational graph data alone. Sparsity assumptions may be unrealistic, and generative models can be computationally expensive. Their primary value may therefore be conceptual and diagnostic: they offer a principled language for separating stable mechanisms from accidental correlations.

11. Source-Centric, Data-Centric, and Source-Free Adaptation

The field has increasingly moved beyond the assumption that the source graph is fixed and always accessible.

11.1. Source Selection

If several source graphs are available, an ideal system should estimate which source is relevant to a given target. A simple source weighting scheme is

$$w_k = \frac{\exp(-d(G_k^s, G_t)/\tau)}{\sum_{r=1}^K \exp(-d(G_r^s, G_t)/\tau)}. \quad (24)$$

The challenge is to choose a distance d that predicts transferability rather than superficial similarity.

11.2. Graph Data Selection

GRADATE reframes GDA as a data-selection problem rather than only a model-learning problem. The method selects source training samples using optimal transport and does not require a particular GNN prediction mechanism, making it complementary to model-centric adaptation [14].

This perspective is important because not all source nodes are equally useful. A source graph may contain regions that are highly relevant to the target and regions that induce negative transfer.

11.3. Multi-Source-Free Adaptation

GraphATA studies the practical setting in which multiple labeled source graphs are no longer accessible. Instead of transferring source data, it aggregates source-model parameters at node granularity according to local context [15].

This shifts the adaptation problem from

$$\{G_1^s, \dots, G_K^s\} \rightarrow G_t \quad (25)$$

to

$$\{f_{\theta_1}, \dots, f_{\theta_K}\} \rightarrow f_{\theta_t}. \quad (26)$$

11.4. Why Source-Free Learning Matters

Source-free adaptation is increasingly relevant when source data contain personal, proprietary, or regulated information. It also changes the optimization landscape: without the original labels, the target graph itself must provide enough information to identify useful combinations of source knowledge.

12. Open-Set Graph Domain Adaptation

Closed-set assumptions are convenient but unrealistic in many deployment settings. Wang et al. formalize open-set GDA for node classification, where the target contains classes not present in the

source [7]. GraphRTA later explores unsupervised open-set GDA through dual reprogramming of the graph and model, together with an explicit unknown category [16].

12.1. Problem Formulation

Let $\mathcal{Y}_s$ be the known source label set. For open-set adaptation,

$$\mathcal{Y}_t = \mathcal{Y}_s \cup \mathcal{Y}_u, \quad \mathcal{Y}_u \cap \mathcal{Y}_s = \emptyset. \quad (27)$$

A target node may therefore belong to a shared class or to an unknown class.

12.2. Why Open-Set GDA Is Harder Than Open-Set Domain Adaptation

Graph structure creates two additional complications. First, unknown nodes can be connected to known nodes, so their neighborhood information can contaminate class prototypes. Second, unknown subgraphs can possess structural patterns that look similar to known categories. Therefore, unknown detection cannot rely only on feature distance.

12.3. Evaluation

Open-set evaluation should separately measure known-class accuracy and unknown-class detection. Useful metrics include macro-F1 over known classes, unknown recall, AUROC for unknown detection, and harmonic means such as

$$H = \frac{2A_{\text{known}}A_{\text{unknown}}}{A_{\text{known}} + A_{\text{unknown}}}. \quad (28)$$

This prevents a system from appearing strong by simply rejecting too many target nodes as unknown.

13. Continual and Evolving Graph Domain Adaptation

Real graph distributions are dynamic. Instead of adapting once, a deployed model can encounter a sequence of target graphs $G_1, G_2, \dots, G_T$. This creates a continual adaptation problem.

13.1. Why Single-Step GDA Is Insufficient

A model optimized for one target can become poorly calibrated on later domains. Updating on every new graph can also cause catastrophic forgetting. Thus a continual learner needs two objectives:

$$\min \mathcal{L}_{\text{new}} + \beta \mathcal{L}_{\text{retain}}. \quad (29)$$

The first term supports adaptation to the current domain; the second preserves transferable information learned previously.

13.2. Graph Continual Adaptive Learning

GCAL addresses evolving domain shifts through a bilevel strategy. Its adaptation phase uses information maximization to update the model on new graph domains while the memory-generation phase compresses previous graphs into memory structures guided by an information-bottleneck-derived objective [13].

13.3. Memory as a First-Class Transfer Object

This development suggests a broader design principle: the transferable knowledge of a graph model need not be stored only in parameters. It can also be stored as representative subgraphs, graph statistics, class prototypes, or compressed memories. Future continual GDA may therefore combine model memory with structural memory.

14. Mathematical Analysis and Generalization

A classic domain-adaptation intuition is that target risk can be bounded by source risk plus a discrepancy between domains and a term describing how different their labeling mechanisms are. A stylized graph-aware bound is

$$R_t(f) \leq R_s(f) + d_{\mathcal{H}}(P_s, P_t) + \lambda^*, \quad (30)$$

where $d_{\mathcal{H}}$ is a task-dependent divergence and λ^* represents the best joint error attainable by a hypothesis that performs well in both domains.

For graphs, the discrepancy should be sensitive to relational structure. One possible decomposition is

$$d(P_s, P_t) = \alpha d_X + \beta d_A + \gamma d_H + \delta d_Y, \quad (31)$$

where d_X measures feature mismatch, d_A structural mismatch, d_H homophily mismatch, and d_Y label-conditional mismatch.

This decomposition is conceptual rather than universal, but it explains an important empirical pattern: reducing one discrepancy term cannot guarantee improved target performance if another term dominates.

14.1. Negative Transfer

Negative transfer occurs when adaptation decreases target performance relative to a source-only baseline. It is particularly likely when source and target have weak task relatedness or conflicting structural semantics.

A practical transferability score can be defined conceptually as

$$T(G_s, G_t) = U_{\text{adapt}}(G_s, G_t) - U_{\text{source-only}}(G_t), \quad (32)$$

where U is a target utility measure. The difficulty is that T cannot be observed directly without target labels. This motivates unsupervised transferability estimation, a major open research problem.

15. Training Objectives and Optimization Patterns

Modern GDA methods often combine several losses. A representative composite objective is

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \lambda_1 \mathcal{L}_{\text{align}} + \lambda_2 \mathcal{L}_{\text{str}} + \lambda_3 \mathcal{L}_{\text{proto}} + \lambda_4 \mathcal{L}_{\text{ent}} + \lambda_5 \mathcal{L}_{\text{cons}}. \quad (33)$$

15.1. Alignment versus Discrimination

A transfer objective must balance two competing goals. Alignment attempts to reduce domain information; discrimination attempts to preserve label information. If the alignment term dominates, classes can collapse. If the task loss dominates, no adaptation may occur.

15.2. Confidence Scheduling

Pseudo-label-based methods often use a curriculum. Early iterations select only high-confidence target nodes. As the model improves, the threshold is relaxed. A generic schedule is

$$\tau_t = \tau_{\min} + (\tau_{\max} - \tau_{\min}) \exp(-kt). \quad (34)$$

Such schedules are heuristics rather than universal solutions, but they emphasize that target-label uncertainty should be modeled explicitly.

16. Evaluation Protocols and Benchmarks

Evaluation methodology is especially important in GDA because apparently small differences in benchmark construction can change the difficulty of transfer.

16.1. Source-Only, Adapted, and Target-Supervised Baselines

Every study should report at least three reference points:

1. Source-only performance.
2. Proposed unsupervised adaptation performance.
3. Target-supervised performance when target labels are available for evaluation or upper-bound analysis.

16.2. Transfer Scenario Design

A realistic benchmark should create shifts that resemble deployment. Examples include cross-network transfer, cross-time transfer, cross-geography transfer, cross-platform transfer, and cross-collection-protocol transfer.

Randomly splitting nodes from a single graph is usually insufficient to represent domain adaptation because the source and target can remain too similar.

16.3. Metrics

For balanced node classification, accuracy is intuitive. Under class imbalance, macro-F1 is often more informative. Domain-level metrics can also be useful, such as a distance between source and target embeddings, but a reduction in embedding discrepancy should never be interpreted as evidence of successful adaptation without a task metric.

16.4. Benchmark Quality Criteria

We recommend five criteria.

Table 4. Recommended criteria for graph domain adaptation benchmarks.

Criterion	Question to answer
Shift authenticity	Does the source-target difference resemble a real deployment shift?
Task relatedness	Are the source and target tasks genuinely related?
Structural variation	Are topology and graph statistics allowed to change?
Label-space trans- parency	Are shared and unknown classes clearly documented?
Reproducibility	Are preprocessing, splits, hyperparameters, and seeds reported?

16.5. Benchmark Families

The literature contains cross-network social or information graphs, graph classification datasets, image graphs, product/review graphs, and evolving graph collections. The 2025 survey and associated paper list provide a useful starting point for compiling published transfer scenarios [1,40].

17. Comparative Taxonomy of Representative Methods

Table 5. Representative graph domain adaptation methods and their primary adaptation mechanisms.

Method	Main idea	Adaptation object	Setting	Primary challenge addressed
CDNE [4]	Cross-network deep embedding	Representation	Single-source node-level	Cross-network transfer
ACDNE [3]	Adversarial embedding alignment	Representation/objective	Single-source node-level	Domain discrepancy
Aligned Graph Bases [2]	Learn coherent graph Fourier bases	Spectrum/structure	Node-level	Spectral mismatch
UDA-GCN	GCN with domain adaptation	Representation/propagation	Node-level	Unsupervised transfer
Open-set GDA [7]	Separate known and unknown alignment	Prediction/objective	Open-set node-level	Unknown target classes
TO-UGDA [5]	Target-oriented adaptation	Objective/target	Node/graph	Target relevance
Propagation Re-thinking [6]	Emphasize propagation	Propagation	Node-level	Structure-induced shift
JDAGCN [8]	Joint domain and structural alignment	Structure/objective	Node-level	Structural mismatch
TDSS [9]	Target structural smoothing	Target graph	Node-level	Local structural shift
Homophily-GDA [10]	Align graph homophily	Homophily/spectrum	Node-level	Homophily discrepancy
DGSDA [12]	Disentangle attribute/topology	Spectrum/structure	Node-level	Attribute-topology entanglement
SLOGAN [11]	Sparse causal discovery + intervention	Causality/generation	Node-level	Spurious correlation
GraphATA [15]	Node-centric source model aggregation	Source model	Multi-source-free	Source unavailability
GRADATE [14]	Optimal-transport data selection	Source data	Graph-level	Severe shifts/data cost

Method	Main idea	Adaptation object	Setting	Primary challenge addressed
GraphRTA [16]	Dual graph/model reprogramming	Target + model	Open-set	Unknown classes
GCAL [13]	Continual adaptation + memory	Model/memory	Continual	Catastrophic forgetting
SATMC [17]	Sequential structure/attribute transformation	Source/target	Node-level	Heterogeneous shifts
PMGCDA [18]	Prototypical multi-source contrastive alignment	Multi-source	Node-level	Source weighting

Interpretation. The table suggests that modern GDA has moved from one-dimensional feature alignment toward joint control of structure, class semantics, source relevance, and temporal memory.

18. Critical Failure Modes

18.1. Over-Alignment

Aligning all target nodes with all source nodes can erase class boundaries. This is most dangerous when label priors differ or the target contains unknown classes.

18.2. Pseudo-Label Confirmation Bias

If an incorrect pseudo-label enters the target prototype, contrastive or prototype alignment can amplify the error. Confidence alone is insufficient because a biased model can be confidently wrong.

18.3. Structural Contamination

When unknown or unrelated nodes are connected to target nodes of interest, message passing can leak misleading information. This makes neighborhood filtering and structural reprogramming attractive in open-set settings.

18.4. Source Dominance

A large source graph can dominate statistics simply because it has more nodes. Multi-source methods therefore need node-level or domain-level weighting [18].

18.5. Benchmark Leakage

Pretraining, hyperparameter tuning, or data preprocessing may inadvertently use target information. This is especially easy in graph settings where connected components or duplicate entities can cross source-target boundaries.

18.6. Architecture-Induced Bias

Adaptation claims can be confounded by improvements in the underlying GNN rather than the domain adaptation component. Strong ablation studies should separate architecture improvements from transfer mechanisms.

19. Practical Design Guide

A practical GDA system can be designed using the following workflow.

Algorithm 1 A generic workflow for unsupervised graph domain adaptation

Require: Labeled source graph(s) G_s , unlabeled target graph G_t

- 1: Train or load a source graph encoder F_θ
 - 2: Measure source-target differences in features, topology, and homophily
 - 3: Select an adaptation mechanism appropriate to the dominant shift
 - 4: Compute source supervised loss $\mathcal{L}_{\text{sup}}$
 - 5: Compute alignment and structural losses
 - 6: Generate target pseudo-labels only above a confidence threshold
 - 7: Update encoder/classifier using the combined objective
 - 8: Evaluate target performance and compare against source-only baseline
 - 9: **if** negative transfer is detected **then**
 - 10: Reduce source weight or increase target-specific adaptation
 - 11: **end if**
 - 12: Repeat until convergence
-

19.1. Choosing the Adaptation Family

If the dominant issue is feature mismatch with similar structure, representation alignment is a sensible starting point. If topology differs strongly, structural or spectral methods are more appropriate. If homophily changes, propagation or homophily-aware adaptation should be prioritized. If multiple source graphs exist, source weighting becomes central. If the source cannot be accessed, source-free methods are required. If target classes are unknown, open-set adaptation must be considered from the beginning.

19.2. Recommended Ablations

At minimum, remove each of the following one at a time: domain alignment, structural alignment, target regularization, source weighting, and pseudo-labeling. This reveals whether the improvement comes from adaptation or simply from a stronger base GNN.

20. Applications

20.1. Social and Information Networks

Cross-network transfer is natural when two platforms share user roles or semantic categories but have different interaction patterns. CDNE and ACDNE were developed in this context [3,4].

20.2. Recommendation

User-item graphs differ across countries, platforms, and product catalogs. Source adaptation can exploit behavior learned in large markets while reducing labeling costs in smaller markets.

20.3. Fraud and Security

Fraud patterns evolve and are often highly imbalanced. Structural shifts can be especially important because attackers modify relational behavior as well as individual features.

20.4. Biomedicine and Chemistry

Molecular and biological graphs may differ across laboratories, assay platforms, or patient populations. GDA can potentially reduce the need for large labeled datasets in new experimental settings.

20.5. Scientific and Industrial Systems

Transportation graphs, sensor interaction graphs, power networks, and industrial dependency networks can differ across locations or operating regimes. Adaptation can help transfer learned mechanisms while preserving target-specific structure.

21. Research Trends Through 2026

Several trends are particularly visible in the recent literature.

21.1. From Marginal Alignment to Structural Alignment

The field has moved from global embedding alignment toward explicit treatment of propagation, topology, homophily, and spectral behavior [6,9,10,12,17].

21.2. From Single-Source to Multi-Source Transfer

Recent methods treat each source as having potentially different utility. Node-level and domain-level transferability weighting are increasingly prominent [15,18].

21.3. From Model-Centric to Data-Centric Adaptation

GRADATE demonstrates that source data selection can itself be the adaptation mechanism and can complement model-centric methods [14].

21.4. From Closed-Set to Open-Set Transfer

Open-set GDA is moving toward explicit unknown detection and graph/model reprogramming rather than thresholding alone [7,16].

21.5. From Static to Continual Adaptation

GCAL reframes domain adaptation as a sequence of evolving shifts rather than a single source-to-target jump [13].

21.6. From Correlation to Causality

SLOGAN reflects an emerging attempt to separate stable causal features from domain-specific spurious correlations [11].

22. Open Problems and Future Directions

22.1. Transferability Estimation

The most important missing primitive is an unsupervised estimate of whether a source is worth transferring. A useful estimator should predict whether adaptation will outperform a source-only model without observing target labels.

22.2. Causal Invariance on Graphs

Future methods should integrate graph causal discovery, invariant risk minimization, and structural interventions. The key question is not merely which embeddings are invariant, but which mechanisms remain valid across environments.

22.3. Adaptive Source Routing

For large collections of source graphs, a single global source weight is unlikely to be optimal. The natural extension is a routing model that chooses source knowledge separately for each target node, subgraph, or task.

22.4. Uncertainty-Aware Adaptation

A robust system should represent uncertainty over pseudo-labels, source relevance, and unknown classes. Bayesian approximations, conformal prediction, or calibrated ensembles could make adaptation safer.

22.5. Large-Scale Structural Matching

Exact graph alignment is often computationally infeasible. Scalable methods need subgraph sampling, approximate optimal transport, graph coarsening, or learned structural sketches.

22.6. Graph Foundation Models

As graph foundation models mature, adaptation may shift from training an entire GNN per domain to learning compact domain adapters, prompts, or low-rank parameter updates. The challenge will be preserving graph-specific inductive bias while allowing rapid specialization.

22.7. Privacy and Federated GDA

Source-free and source-model approaches naturally connect to federated settings where raw source graphs cannot be shared. Future work should jointly optimize adaptation quality and privacy guarantees.

22.8. Benchmark Standardization

A mature benchmark suite should report feature shift, structure shift, homophily shift, class-prior shift, and label-space shift separately. This would allow researchers to answer a more meaningful question: which methods work under which kinds of graph shift?

23. Conclusion

Graph domain adaptation has evolved from a relatively narrow problem of cross-network representation transfer into a broader study of how relational knowledge should move across environments. Early work established that graph-specific spectral or embedding structure can be transferred across networks [2–4]. More recent research demonstrates that successful adaptation must account for propagation, homophily, structural heterogeneity, target uncertainty, causal mechanisms, source relevance, open-set classes, and continual drift [6,10,11,13,18].

The most useful conceptual shift is therefore from *domain matching* to *transferable knowledge identification*. A source and a target graph should not necessarily be made identical. Instead, an adaptation system should discover which attributes, structures, frequencies, prototypes, causal factors, and model parameters are stable enough to transfer and which are domain-specific enough to suppress.

This perspective also clarifies why the next generation of GDA methods will likely be more adaptive, selective, and uncertainty-aware. Structural alignment, source routing, causal disentanglement, data selection, source-free learning, and continual memory are all different answers to the same underlying question: *what should be transferred, to whom, and under which conditions?* A general solution to this question would transform GDA from a collection of alignment heuristics into a principled framework for robust graph learning under distribution shift.

Checklist for Future GDA Studies

- ☐ Define source and target graphs explicitly.
- ☐ Report whether the setting is closed-set, open-set, source-free, or multi-source.
- ☐ Quantify feature, structure, homophily, and class-prior shifts.
- ☐ Include a source-only baseline.
- ☐ Separate architecture gains from adaptation gains.
- ☐ Report multiple random seeds.
- ☐ Use target labels only for evaluation, not tuning, unless the setting explicitly permits validation.
- ☐ Report both task metrics and adaptation diagnostics.

- ☐ Analyze negative transfer cases, not only average gains.
☐ Release splits, preprocessing, hyperparameters, and code when possible.

Appendix X Extended Taxonomy and Reporting Template

Appendix X.1 Two-Axis Taxonomy

The following matrix places representative approaches on the two dimensions emphasized throughout this survey: the *object of adaptation* and the *location of intervention*.

Table A6. Two-axis taxonomy for graph domain adaptation.

Adaptation object	Source-side	Adaptation-objective side	Target-side
Representation	Source feature filtering, source data selection, source model aggregation	MMD, adversarial loss, class-conditional alignment, contrastive loss	Target normalization, target re-embedding, structural smoothing
Structure	Graph rewiring, source structural sampling	Structural discrepancy, topology matching, graph kernels	Target reprogramming, edge filtering, neighborhood adaptation
Signal / spectrum	Source spectral compression	Spectral filter matching, frequency alignment	Target spectral filtering and smoothing
Prediction	Source prototype construction	Entropy minimization, pseudo-label consistency	Unknown-class detection, target confidence calibration
Causal factors	Source invariant-factor extraction	Mutual-information bottlenecks, intervention objectives	Target intervention or factor reconstruction
Memory	Source memory generation	Continual retention objectives	Target-aware memory retrieval and replay

Appendix X.2 Recommended Reporting Template

A reproducible GDA paper should report the following fields for every experiment.

Table A7. Recommended minimum reporting template for graph domain adaptation experiments.

Field	Recommended content
Source / target definition	Exact graph datasets, graph construction procedure, node and edge semantics.
Task	Node, edge, graph, or graph-set prediction; number of classes.
Shift description	Feature, structure, homophily, label-prior, temporal, geographic, or label-space shift.
Label availability	Source labels, target labels for evaluation only, validation protocol.
Architecture	GNN family, layers, hidden size, normalization, aggregation function.
Adaptation mechanism	Alignment, structural adaptation, prototype learning, causal intervention, source selection, or memory.
Hyperparameters	Adaptation weights, thresholds, temperature, learning rate, epochs.
Randomness	Number of runs, seeds, mean, standard deviation or confidence intervals.
Baselines	Source-only, target-supervised where permitted, and representative modern GDA baselines.
Failure analysis	Negative transfer cases, unknown classes, structural mismatch, calibration.
Efficiency	Runtime, memory, number of source graphs, number of nodes/edges used.
Reproducibility	Code, preprocessing scripts, splits, configuration files, and environment.

709 *Appendix X.3 Glossary***Table A8.** Glossary of recurring GDA terminology.

Term	Meaning
GDA	Graph domain adaptation: transferring knowledge between graph domains under distribution shift.
UGDA	Unsupervised graph domain adaptation with unlabeled target graphs.
Source-free GDA	Adaptation when original source graphs and labels are unavailable at target-time training.
Multi-source GDA	Transfer from multiple labeled or pretrained source domains.
Open-set GDA	Target label space contains classes absent from the source.
Homophily	Tendency of linked nodes to share labels or attributes.
Structural shift	Difference in topology or structural statistics across domains.
Pseudo-label	A model-generated target label used as a surrogate for supervision.
Prototype	Representative class centroid or summary embedding.
Negative transfer	Target performance degradation caused by adaptation.
Transferability	Degree to which a source provides useful information for a given target.
Continual adaptation	Sequential adaptation to evolving domains while retaining prior knowledge.
Domain-private information	Information useful within one domain but not reliably transferable.
Domain-invariant information	Information whose predictive relationship remains stable across domains.

710 *Appendix X.4 Suggested Research Questions*

711 The following questions define a useful experimental agenda for future work:

- 712 1. Can a model estimate transferability without target labels?
- 713 2. Which structural statistics best predict cross-domain GNN degradation?
- 714 3. When is homophily alignment more useful than feature alignment?
- 715 4. Can source-free adaptation preserve privacy without causing severe performance loss?
- 716 5. How should an adapter detect that a target node belongs to an unseen class?

6. Can a graph foundation model learn a reusable notion of structural invariance across domains?
7. How should adaptation be evaluated when the target graph changes continuously?

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